

| Water Source HP Outdoor Unit          |              |           |                                                    |                                                                                                                |
|---------------------------------------|--------------|-----------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Component                             | Abbreviation |           | Cooling Mode                                       |                                                                                                                |
| Compressor                            | Comp         | Operation | Target low pressure control (default: 113.8psig)   | Primary function, Maintain Average Evaporator Temperature (Average temperature of Eva. In Sensors)             |
|                                       |              | Range     | 14~120Hz                                           | Low pressure range 100psig - 128 psig. Evaporator temperature range 45 °F -48°F                                |
| High side pressure                    | HP           | Operation | Head pressure is effected by the water temperature | Is there a variable flow control valve installed? This will effect head pressure.                              |
|                                       |              | Range     | 299 psig - 469 psig (water temperature 68°F-104°F) | At 228psig, low limit control is activated at 512psig high limit control is activated                          |
| Main EEV                              | E_M          | Operation | Full Open                                          |                                                                                                                |
|                                       |              | Range     | 2000 Steps                                         |                                                                                                                |
| EVI EEV                               | E_EVI        | Operation | *Subcooling control (default: SC=36°F)             | SC = High Side Saturation Temp. - Liquid Line Temp.                                                            |
|                                       |              | Range     | 0-480step                                          |                                                                                                                |
| 4 Way Valve                           | V_4W         | Operation | Off                                                |                                                                                                                |
| EVI By-Pass Solenoid Valve            | V_EB         | Operation | On (Open): *T_W > 104°F or 50Hz↓                   | Activates Sub-Cooling Control                                                                                  |
|                                       |              |           | Off (Close): T_W < 100.4°F↓ and 65 Hz↑             | Discharge Temperature Protection, Control by unit                                                              |
| EVI Solenoid Valve                    | V_EVI        | Operation | On (Open)                                          |                                                                                                                |
| Hot Gas By-Pass                       | V_HG         | Operation | Off (Closed)                                       |                                                                                                                |
| Accumulator Oil Return Solenoid Valve | V_AR         | Operation | On (Open when compressor is running)               | <b>Off</b> during start up if DSH is < 18°F. <b>Off</b> for 2 min. if DSH is < 18°F <b>On</b> when DCSH > 27°F |
| Hot Gas By-Pass 2 (HR Only)           | V_HG2        | Operation | Off Closed                                         | Opens after 20 minutes of continuous run time in cooling mode                                                  |
| Main EEV Solenoid Valve (HR Only)     | V_ME         | Operation | On (Open)                                          | Energized in cooling only                                                                                      |
| Main Cooling Solenoid                 | VMC          | Operation | Off (Closed)                                       |                                                                                                                |
| Liquid Pipe Thermistor                | T_L          | Operation | Used to calculate sub-cooling                      | Sub-Cooling range is 9°F-36°F                                                                                  |
| HP - HR Ball Valve (3 phase only)     | BV_HP        | Operation | Open                                               | Valve must be open for heat pump applications                                                                  |
| IDU                                   |              |           |                                                    |                                                                                                                |
| Component                             | Abbreviation |           | Cooling Mode                                       | Comments                                                                                                       |
| Room Temperature                      | Room Temp.   | Operation | 65~86°F                                            |                                                                                                                |
| Liquid Pipe Thermistor                | EVA In 1     | Operation | Evaporator Saturation Temperature. Range 45-48°F   | Used to calculate Evaporator Superheat                                                                         |
| Gas Pipe Thermistor                   | EVA Out 1    | Operation | Measures Suction Pipe Temperature                  | Used to calculate Evaporator Superheat                                                                         |
| EEV                                   | EEV 1        | Operation | Modulates to maintain Superheat of 3.6 degrees F   | Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)                                                  |
|                                       |              | Range     | 0 - 2000 Steps (0 Steps when zone is satisfied)    | Typical range between 250 and 1500 Steps                                                                       |
| IDU Fan                               | Fan Speed    | Range     | Low , Medium, High, Auto                           |                                                                                                                |
|                                       |              | Operation | Low Speed: Ts - Ti  ≤ 1.8°F                        | TS - Set Temperature                                                                                           |
|                                       |              |           | Medium Speed: 1.8°F <  Ts - Ti  ≤ 5.4°F            | Ti Indoor Temperature                                                                                          |
|                                       |              |           | High Speed 5.4°F <  Ts - Ti                        |                                                                                                                |
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